TNE20003 – Internet and Cybersecurity for Engineering Applications

A

Aims:

- Explain & understand NAT at deeper level

Preparation:

- View "NAT & DHCP"

Due Date:

- Your Task will be assessed via an online quiz. You must score the required minimum to pass the test. You will be allowed a number of attempts to pass the test at the grade level you attempt. You are encouraged to complete the test at the end of the lab but if you do not, you must complete it before your next lab class.

  .

Task 1.

Build the network provided in figure 1 with Cisco Packet Tracer

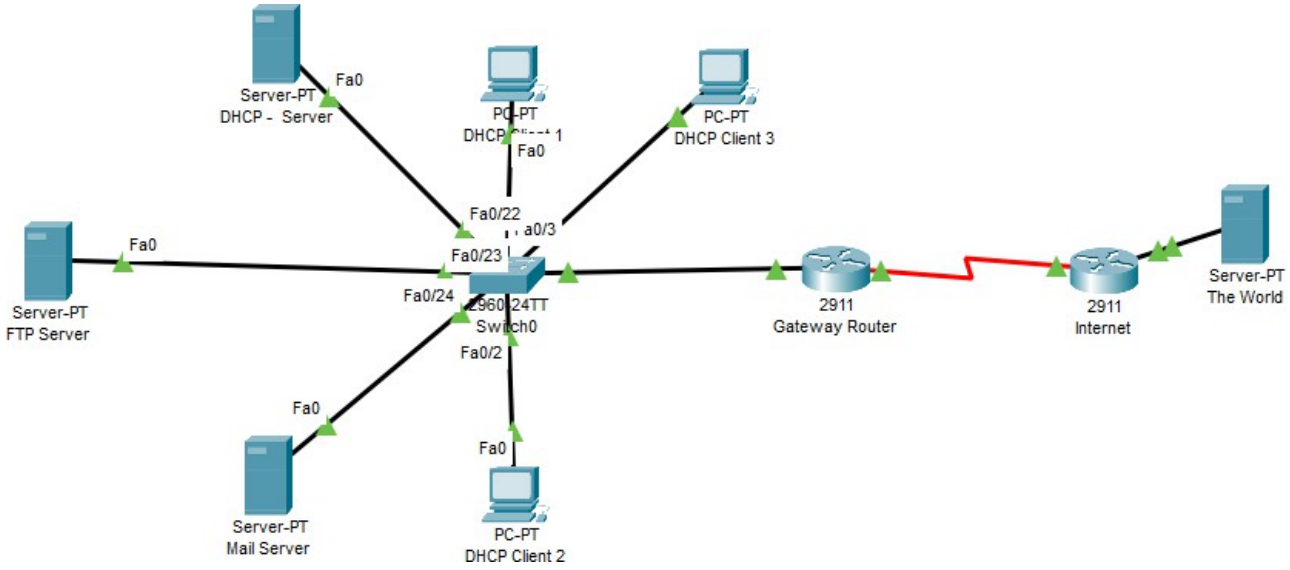


Figure 1

To attempt the D task you must have completed the P and C tasks. Use the network you built and tested in the Credit task for lab 5 to test and answer the following questions.

Task 2.

1. You have implemented NAT successfully in the Pass task.

   a. Now generate some traffic by pinging from DHCP Client 3 to **The World** server located at 136.186.100.2. What happens?

   The ping is successfully working by replies from world server.

   b. Are there any address translations on the **Gateway** Router# prompt? Explain

   There is an address translation on the gateway router because the connection is reached to the world server.

   c. Ping **The World** server located at 136.186.100.2 from the **ftp** server. What happens?
   The ping shown request timed out.

d. How far does the ping from the **ftp** server propagate through the network?
The ping could go from local network to gateway router.

e. Were there any translations on the Gateway router? Explain why/why not.
There is no translation on the gateway way router because the connection is not established to world server.

f. Change the IP address of DHCP Client 3 to a static address of 192.168.1.80. Ping **The World** server located at 136.186.100.2 from this **PC**. What happens?
The ping is successfully reach to the world server.

g. Part of your output should be as shown in the picture below:

```
Router# sh ip nat tran
Pro  Inside global     Inside local       Outside local      Outside global
icmp 200.8.8.1:5       192.168.1.80:5     136.186.100.2:5    136.186.100.2:5
```

h. What does the **:5** represent? The translation port.

i. Now ping from DHCP Client 2 and see the translations. What do you see?

Client 2 Ip address is added to NAT table.

j. What does inside local mean?

The private Ip address that used in the local network.

k. Change the IP address of DHCP Client 3 to a static address of 192.168.1.180. Ping **The World** server located at 136.186.100.2 from this **PC**. What happens?

The ping was requested time out.

l. Were there any translations on the **Gateway** router? Explain why/why not.

No translation on the gateway router because the connection to not establish to the internet or world server.

~~~~~ End of Lab ~~~~~